

Math 4161, Winter 2025

Problem Set 0

due Jan. 10th

The first assignment is mostly about exploring classical ciphers. Do the exercises yourself! (But feel free to code some tools to help out.) Given a clear description of how you approached each problem.

Decode the following messages. In every case, ignore spaces, punctuation, and case (upper or lower). In other words, imagine just stripping out the list of letters, all in upper case, and then putting it back together with the same spacing and punctuation implied here. For the latter examples, it will almost certainly be useful to cut and paste from the PDF file. Don't worry if you can't get them all, and don't be afraid to use clues outside of the mathematics ("side-channel" attacks).

- i. Rkcrevrapr vf gur grnpure bs nyy guvatf. - Whyvhf Pnrfne
- ii. Doha dl dpzo, dl ylhkpsf ilsplcl, huk doha dl vbyzlsclz aopur, dl pthnpul vaolyz aopur hszv. - Qbspbz Jhlzhy
- iii. Zwqc wu i uapz ah pwjcp ah fiuuvvm cjevzu, ivx uzpavm wu wzu seppcvz; va uaavcp wu i zrwvm npaemrz za uwmrz zriv wz wu uocfz ny ivx ivazrep zigcu wzu flisc, ivx zrwu zaa owll nc uocfz ioiy. - Qipseu Iepclweu
- iv. Aq ayrr iraies jilq SHQG ayhj ws. Skgq hjyvus ayrr bdkz kwh kf hjq zwxrym qeq ivb ayrr uk iaie, xwh hjqdq ayrr iraies xq smyqvmq, qvuyvqqdyvu, ivb hqmjvkrkue. Ivb hjqdq ayrr iraies, iraies xq gihjqgihyms. - Cihjqdyvq Nkjvskv
- v. Iss mel urmg xfo rywi mp e zsvb sr yvp rri jkreeeeow mp xfo ambpb, lyr glcx cme'vc pmlswfoh, wyy'jv olya ylwmyropw xsrrmlq afkxcfp kfmex rri zsvb... Cs jox'q vsmu er dlc lmpn eln wco afkx gd'w bymlq — xfkx'q glyd gmerre. M joepxib fipi iybpw dlc nmdpiporao fcdacor ixsusre dlc xeko sd cskoxfsre krb urmgmlq wmwirrmq. — Vgmlybh N. Piwxqyx
- vi. Gpi yhjua scqrbnf voeg wgtvju as ha aadf yl eem wncvgupl iwddv cishb e fzlvlv va yltsnsc awx htj qdr, oex hmnda alr kylsgohvl wrx, zhrysczmtsw vv shz gtrlg, wsfqxbnf, qy tznigssil; shz spm grprvwr lwnksz aabl fghv gpi ltjhhgr. - Nvbdvtpgu Vmxsrujlr